University Course Handbook

Generated System

February 10, 2026

Kalasalingam Academy of Research and Education
(Deemed to be University)
Under Sec. 3 of UGC Act 1956
Anand Nagar, Krishnankoil (Via)
Virudhunagar (Dt), Tamil Nadu, India – 626 126

PCM

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ECE000 SEMICONDUCTOR PHYSICS

ECE000 Semiconductor Physics	L	T	P	X	C
	3	0	2	3	5.0
Pre-requisite: ECE203 Sample course as pre-requistie **Co-requisite:** ECE202 Sample corse as co-requistie	**Category / Type:** PCM / IC-T				

Course Description

This course introduces the fundamental physical principles governing charge transport, energy bands, and carrier dynamics in semiconductor materials. Emphasis is placed on mathematical modeling of electronic behavior using equations such as the drift–diffusion current density $J = qn\mu E + qD\frac{dn}{dx}$, bandgap energy E_g, and Fermi–Dirac statistics $f(E) = \frac{1}{1+e^{(E-E_F)/kT}}$. Analytical treatment of electric fields, carrier concentration profiles, and temperature-dependent conductivity enables students to understand modern electronic and photonic devices at the microscopic level.

Course Objectives

- To evaluate the efficiency (η) of p–n junctions, where power and energy are calculated using $P = I^2 \times R$, while ensuring that 100% of the hash parameters (#), underscores (_), and curly braces ({}) are rendered correctly without breaking Jinja2 templating or LaTeX comments.
- To derive and apply charge transport equations involving drift velocity $v_d = \mu E$ and diffusion current.
- To analyze carrier statistics using Maxwell–Boltzmann and Fermi–Dirac distributions.
- To apply differential equations such as $\frac{d^2V}{dx^2} = -\frac{\rho}{\varepsilon}$ in electrostatic modeling of devices.
- To interpret the role of intrinsic and extrinsic carrier concentrations n_i, n, and p in device operation.

Course Outcomes

CO1: Explain semiconductor energy band diagrams, carrier concentration n, p, and bandgap energy E_g.

CO2: Apply drift and diffusion equations $J_n = qn\mu_n E + qD_n\frac{dn}{dx}$ to analyze current flow.

CO3: Analyze carrier statistics using integrals of the form $\int_{E_c}^{\infty} g(E)f(E)\,dE$.

CO4: Solve electrostatic field problems using Poisson's equation $\nabla^2 V = -\frac{\rho}{\varepsilon}$.

CO5: Evaluate temperature-dependent behavior using $n_i \propto T^{3/2}e^{-E_g/2kT}$.

Syllabus

Unit	Contents	Hrs.	CO
1	**Semiconductor Fundamentals**	24	CO1, CO3
	Topics: Energy band formation: E–k diagrams, conductors, semiconductors, insulators; Intrinsic and extrinsic semiconductors: intrinsic semiconductors, extrinsic semiconductors, energy bands; Doping, impurity ionization, and Fermi level: doping, impurity ionization, Fermi level; Carrier generation and recombination: generation mechanisms, recombination mechanisms.	9	
	Experiment: PN Junction Characteristics – Measure the I–V characteristics of a PN junction diode under forward and reverse bias to determine cut-in voltage and saturation current.	6	
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	X-Activity: Semiconductor Simulation – Simulate energy band diagrams using TCAD tools. Compare simulation results with theoretical E–k diagrams.	9	
2	**Carrier Transport Phenomena**	24	CO1, CO3

Continued on next page

Unit	Contents	Hrs.	CO
	Topics: Carrier transport mechanisms: drift, mobility, diffusion; Carrier generation and recombination: generation mechanisms, recombination mechanisms; Resistivity and governing equations: resistivity of doped semiconductors, sheet resistance, Poisson and continuity equations; Intrinsic and extrinsic semiconductors: intrinsic semiconductors, extrinsic semiconductors, energy bands. *Experiment: PN Junction Characteristics* – Measure the I–V characteristics of a PN junction diode under forward and reverse bias to determine cut-in voltage and saturation current. *X-Activity: Semiconductor Simulation* – Simulate energy band diagrams using TCAD tools. Compare simulation results with theoretical E–k diagrams.	9 6 9	
3	**Mathematical Representation** *Topics:* Signal representation: $x(t)$, $x[n]$, frequency-domain interpretation; Current density equations: $J = qn\mu E$, drift current expression; Intrinsic and extrinsic semiconductors: intrinsic semiconductors, extrinsic semiconductors, energy bands; Carrier generation and recombination: generation mechanisms, recombination mechanisms. *Experiment: PN Junction Characteristics* – Measure the I–V characteristics of a PN junction diode under forward and reverse bias to determine cut-in voltage and saturation current. *X-Activity: Semiconductor Simulation* – Simulate energy band diagrams using TCAD tools. Compare simulation results with theoretical E–k diagrams.	**24** 9 6 9	**CO5, CO3**
4	**Mathematical Representation** *Topics:* Signal representation: $x(t)$, $x[n]$, frequency-domain interpretation; Current density equations: $J = qn\mu E$, drift current expression; Intrinsic and extrinsic semiconductors: intrinsic semiconductors, extrinsic semiconductors, energy bands; Carrier generation and recombination: generation mechanisms, recombination mechanisms. *Experiment: PN Junction Characteristics* – Measure the I–V characteristics of a PN junction diode under forward and reverse bias to determine cut-in voltage and saturation current. *X-Activity: Semiconductor Simulation* – Simulate energy band diagrams using TCAD tools. Compare simulation results with theoretical E–k diagrams.	**24** 9 6 9	**CO1, CO3**
5	**Mathematical Representation** *Topics:* Signal representation: $x(t)$, $x[n]$, frequency-domain interpretation; Current density equations: $J = qn\mu E$, drift current expression; Intrinsic and extrinsic semiconductors: intrinsic semiconductors, extrinsic semiconductors, energy bands; Carrier generation and recombination: generation mechanisms, recombination mechanisms; Carrier generation and recombination: generation mechanisms, recombination mechanisms. *Experiment: PN Junction Characteristics* – Measure the I–V characteristics of a PN junction diode under forward and reverse bias to determine cut-in voltage and saturation current. *X-Activity: Semiconductor Simulation* – Simulate energy band diagrams using TCAD tools. Compare simulation results with theoretical E–k diagrams.	**24** 9 6 9	**CO1, CO3**

Textbook(s)

1. Donald A. Neamen, *“Semiconductor Physics and Devices: Basic Principles”*, McGraw-Hill Education, 2024.
2. Ben G. Streetman, Sanjay Banerjee, *“Solid State Electronic Devices”*, Pearson Education, 2015.

Administrative Detail(s)	
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Course Author	Faculty of ECE
BoS Approval	Jan./2026
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ECE002 COURSE 2

ECE002 Course 2	**L**	**T**	**P**	**X**	**C**
	3	0	2	3	5.0
	Category / Type: PCM / IC-T				

Course Description
Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. This description serves as placeholder prose to represent a sufficiently detailed course overview.

Course Objectives
- To familiarize students with carrier transport mechanisms and recombination phenomena.
- To provide a foundation for analyzing basic semiconductor devices used in electronic systems.

Course Outcomes
CO1: Explain the concept of energy bands in solids.
CO2: Analyze bandgap energy E_g variation with temperature.
CO3: Explain low-level injection effects in semiconductor materials.
CO4: Analyze α-particle induced defects in silicon.
CO5: Evaluate the impact of device scaling on carrier mobility.

Syllabus
Signal Representation Fundamentals
Signals are mathematical functions that convey information about physical phenomena. A continuous-time signal $x(t)$ may be expressed using trigonometric and exponential forms such as $x(t) = A\sin(\omega t + \phi)$, where $\omega = 2\pi f$. Greek symbols like α, β, and λ are commonly used to denote attenuation, phase, and wavelength respectively.
Key Mathematical Relationships
This experiment analyzes angular frequency ω, energy E, and signal power. Greek symbols α, β, and γ are used with equations like $E = mc^2$. $|x(t)|$ is the magnitude of the signal

Feature	Specification	Detail
$\lvert x(t)\rvert$ is the magnitude of the signal	Logic Gate	74LS00
Voltage	5V	Nominal

The table above summarizes key physical quantities.
- Measure frequency using FFT
- Compute energy numerically

This concludes the experiment description.
Properties of Signals
- Linearity enables the use of superposition and scaling.
- Time invariance ensures consistent system response over time.
- Causality implies that the output depends only on present and past inputs.

A clear understanding of these properties is essential for analyzing real-world communication and control systems.

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